# Problem A. Cipher

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|---|---|
| Input file: | `standard input` |
| Output file: | `standard output` |
| Time limit: | 2 seconds |
| Memory limit: | 1024 megabytes |

Alice designed a cipher system: it encodes the plaintext as a 64-bit unsigned integer $x$, and randomly selects a 64-bit unsigned integer $a$ as the public key, encrypting the plaintext to get ciphertext $b = a^x \bmod 2^{64}$. In particular, it is defined that $0^0 = 1$.

Now, Bob has intercepted $n$ sets of encrypted information $(a_i, b_i)$ $(i = 1, 2, \cdots, n)$. To decrypt all the information, Bob needs to find the **smallest** integer $x_i$ $(0 \le x_i < 2^{64})$ for each set of encrypted information such that ${a_i}^{x_i} \equiv b_i \pmod{2^{64}}$, or determine that the information has been corrupted (i.e., there is no $x_i$ that satisfies the condition). Please write a program to help Bob accomplish this task.

## Input

The first line of input contains an integer $n$ $(1 \le n \le 10^5)$, representing the number of messages.

The next $n$ lines, the $i$-th line contains two integers $a_i$ and $b_i$ $(0 \le a_i, b_i < 2^{64})$, representing the public key and ciphertext of the $i$-th set of information, respectively.

## Output

Output $n$ lines, for the $i$-th line:

- If there exists at least one $x_i$ that satisfies the conditions stated in the problem, output the smallest one;
- Otherwise, output a line `broken message`.

## Example

| standard input | standard output |
|---|---|
| 5<br>2 4<br>3 27<br>1000000007 998244353<br>4 2<br>1000000007 1000000009 | 2<br>3<br>994996658310742016<br>broken message<br>broken message |